

Solução do Problema Bandm1 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: Bandm1
- Número de Variáveis: 472
- Número de Restrições: 305
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: -1.586e+02
- Valor Dual: -1.586e+02
- Gap: -6.193e-18
- Número de Iterações: 17

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z)
com valores ≥ 0 do problema Bandm1

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
4	3.420e-01	3	4.019e-01
8	1.283e+01	5	1.949e-01
11	1.446e+00	6	2.862e-01
14	4.613e-01	7	5.377e-02
16	1.240e+00	9	5.481e-01
19	1.192e+00	10	4.679e-01
21	3.379e+00	12	1.854e-01
25	6.756e-01	13	2.631e+00
26	9.093e-02	15	2.100e-01
44	1.729e-01	17	5.158e+00
63	2.475e+00	18	1.580e-01
65	1.882e+01	20	4.875e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
68	1.359e-01	24	5.000e+00
70	9.785e-01	27	1.819e+01
71	2.899e+00	28	9.758e+00
72	3.364e+00	29	9.087e+00
74	3.277e+00	30	9.711e-01
75	7.137e-01	31	1.000e+00
76	8.628e-01	32	2.819e+00
77	1.092e+00	33	2.007e+00
78	2.150e+00	34	2.966e-02
79	1.738e-02	35	2.733e+00
80	2.253e+00	36	2.320e+00
81	1.469e+00	37	3.082e+00
82	1.812e+00	38	3.835e-02
83	7.990e+00	39	3.912e+00
85	3.080e-01	40	2.520e+00
86	3.508e-01	41	3.734e+00
87	5.150e-01	42	3.130e+00
89	1.346e+00	43	1.674e-02
91	1.980e+00	45	3.841e-02
93	5.174e-01	46	4.542e-02
94	9.421e-01	47	4.159e-02
95	1.762e-01	48	3.870e+00
96	1.152e+00	49	3.719e+00
97	3.077e+00	50	3.677e-02
100	1.787e+01	51	3.843e+00
101	1.251e+01	52	4.048e-02
102	2.217e+01	53	4.226e-02
103	1.021e+01	54	4.524e-02
104	2.339e+00	55	4.503e+00
105	4.949e+01	56	5.085e-02
106	1.365e+01	57	6.160e-02
107	6.118e+00	58	4.282e-02
108	2.011e+01	59	4.229e-02
112	3.641e+00	60	5.377e+00
114	1.742e+01	61	4.072e+00
115	5.135e+00	62	2.167e+00
117	1.764e+01	64	2.865e-01
122	9.901e+00	66	1.256e+00
124	1.369e-01	67	7.086e-01
125	1.424e+01	69	1.455e-01
126	4.849e+01	73	2.886e-02
127	6.452e+00	84	4.893e-02
129	1.419e+01	88	2.804e-01
130	6.388e+00	90	1.282e+00
132	1.262e+01	92	6.609e-02
139	1.686e+01	98	1.582e-01
140	3.225e+00	99	1.200e-01
141	2.339e+00	109	5.203e-02
143	6.981e-02	110	1.462e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
146	6.093e+00	111	1.090e-02
147	4.331e+01	113	6.200e-02
148	1.390e+00	116	5.861e-02
149	6.392e-02	118	6.088e-04
150	5.741e+01	119	1.072e-02
151	4.298e+00	120	2.067e-01
153	1.450e-01	121	7.562e-02
154	9.300e-01	128	5.409e-02
155	7.627e+00	133	1.911e-01
156	1.936e+02	134	7.162e-02
157	2.591e+00	135	7.090e-02
158	1.811e+00	136	2.527e-02
159	1.302e+01	137	9.977e-03
160	2.873e+00	138	3.587e-01
161	1.711e-01	142	2.445e-01
162	1.534e+01	144	2.844e-02
163	2.598e+00	145	6.465e-03
164	4.792e+00	152	2.903e-01
165	3.561e-01	167	9.770e-03
166	1.686e+01	169	4.891e-01
168	2.039e+00	174	1.755e-01
171	1.830e+00	175	4.871e-01
172	2.974e-01	177	3.520e-02
173	1.035e+00	178	3.290e-02
176	9.033e+00	180	6.493e-02
179	3.423e+00	181	2.196e-01
182	1.281e+00	183	3.246e-02
185	5.291e-01	184	3.492e-01
186	5.780e-01	192	3.258e-02
187	6.400e-01	193	2.947e-01
188	6.362e+00	199	5.471e-02
189	1.588e+00	200	7.724e-01
190	2.995e+00	203	7.942e-01
191	1.896e+00	204	1.375e-01
194	1.379e-01	206	1.276e-02
195	1.302e-01	210	4.180e-01
196	3.399e-01	213	4.547e-02
197	1.764e+00	214	2.503e-01
198	9.534e-02	215	3.124e-01
201	7.305e-03	216	5.768e-01
202	2.000e+00	219	7.540e-02
205	4.460e-01	221	7.472e-03
207	2.050e-01	222	1.153e-02
208	2.748e-01	225	3.539e-03
209	2.380e-02	227	3.007e-02
211	1.896e-01	228	7.631e-02
212	7.881e-02	229	3.870e-02
217	1.021e-01	233	2.984e-02
218	5.919e-01	234	2.021e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
220	4.082e+00	236	3.242e-02
223	1.856e+01	237	5.482e-02
224	1.856e+01	238	9.385e-02
230	1.764e+01	240	6.106e-02
231	3.796e-01	246	1.000e-01
232	1.424e+01	247	1.877e-01
235	1.437e+01	248	7.635e-01
239	6.388e+00	255	2.992e-01
241	6.388e+00	259	9.899e-03
242	9.326e-02	263	6.114e-02
243	2.580e+00	264	3.803e-01
244	1.431e+00	267	2.659e+00
245	1.279e+00	268	2.838e+00
249	1.224e+00	269	2.633e+00
250	6.014e-01	270	2.776e+00
251	1.378e+00	271	2.840e+00
252	3.500e-01	272	2.683e+00
253	6.870e-01	301	1.000e-04
254	1.600e+00	311	7.595e-01
256	3.485e-01	318	8.480e-01
257	6.628e-01	326	5.828e-02
258	3.545e+00	327	6.186e-01
260	1.758e+00	338	2.114e-02
261	8.115e-01	341	5.000e-03
262	8.337e-01	351	2.000e-03
265	7.400e+00	354	2.273e-01
273	1.636e+00	355	4.086e-01
274	5.024e+00	356	1.972e-01
275	9.810e-01	360	1.730e-03
276	2.726e+00	363	3.302e-01
277	2.042e+00	364	1.397e-01
278	1.700e+00	365	1.640e-01
279	1.007e+02	374	2.438e+00
280	1.535e+02	382	9.094e-02
281	5.035e-02	383	1.984e-01
282	7.800e-02	385	7.040e-01
283	4.973e+01	386	2.875e+00
284	9.793e+02	389	4.476e-01
285	3.266e+02	391	1.059e-03
286	3.266e-01	392	1.858e-02
287	1.960e+02	393	5.690e-02
288	1.062e+01	394	1.771e-01
289	2.125e+01	400	1.402e-01
290	3.601e+01	409	1.989e-02
291	2.068e+02	413	4.502e-02
292	2.068e+02	414	4.862e-02
293	1.034e+03	415	1.090e+00
294	1.570e-01	416	3.990e-01
295	1.363e+00	418	6.865e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
296	1.871e-02	420	2.328e-01
297	4.040e+00	425	3.391e-02
298	2.020e+01	437	1.339e+00
299	1.525e+01	439	2.953e-01
300	1.247e+01	441	1.481e-02
302	6.954e+00	442	1.942e-02
303	2.027e+01	444	1.898e-01
304	2.696e+01	446	6.643e-01
305	3.792e-02	459	1.000e-02
306	1.517e+01		
307	4.281e+00		
308	1.642e+00		
309	2.162e+01		
310	2.162e+01		
312	3.281e+00		
313	1.437e+00		
314	6.532e+01		
315	5.317e+00		
316	1.147e+00		
317	2.240e-01		
319	1.946e+01		
320	7.630e-01		
321	2.830e-01		
322	1.000e+00		
323	3.940e-01		
324	1.530e-01		
325	9.951e+00		
328	2.770e+01		
329	3.694e+00		
330	2.844e+00		
331	8.080e+00		
332	4.250e+00		
333	2.183e+00		
334	5.073e-01		
335	1.369e+00		
336	5.207e+00		
337	2.828e+01		
339	4.406e+00		
340	4.278e+00		
342	1.889e+00		
343	8.639e+00		
344	3.510e-01		
345	8.672e+00		
346	9.501e-01		
347	3.428e+00		
348	7.585e+00		
349	8.561e+00		
350	5.472e+00		
352	2.820e+00		

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
353	1.063e+00		
357	5.352e+00		
358	1.937e+00		
359	1.295e+00		
361	1.372e+00		
362	1.778e+00		
366	6.031e-01		
367	4.987e-01		
368	2.962e+00		
369	8.850e-01		
370	3.451e-01		
371	1.019e+00		
372	2.317e+00		
373	1.544e+00		
375	1.394e+00		
376	5.654e+00		
377	1.655e+00		
378	3.640e-02		
379	1.499e+00		
380	2.209e+00		
381	2.880e+00		
384	3.230e+00		
387	2.706e-01		
388	6.661e-01		
390	6.160e+00		
395	4.451e+00		
396	9.288e-01		
397	2.549e+00		
398	2.612e+00		
399	1.000e+00		
402	3.738e+00		
403	1.890e+00		
404	5.180e-01		
405	2.280e-01		
406	1.380e-01		
407	6.649e-01		
408	4.580e+00		
410	1.984e+00		
411	1.456e+00		
412	5.688e+00		
417	9.656e-02		
419	1.780e+00		
421	1.577e+00		
422	3.486e-01		
423	1.136e-01		
424	1.776e+00		
426	6.968e-01		
427	5.348e-01		
428	1.986e-01		

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
429	2.900e-01		
430	3.740e-01		
431	1.319e+01		
432	1.750e+00		
433	1.427e+00		
434	6.362e+00		
435	6.509e+00		
436	2.074e+00		
438	2.000e+00		
440	9.465e+00		
443	8.768e-02		
445	5.014e+00		
447	1.560e-01		
448	1.282e+00		
449	1.058e+00		
450	1.903e+00		
451	4.210e-01		
452	2.654e-01		
453	1.153e-01		
454	3.476e+00		
455	2.464e+01		
456	2.681e+00		
457	4.218e-02		
458	3.309e+00		
460	4.800e-01		
461	3.388e+00		
462	2.230e-01		
463	3.450e-01		
464	2.793e+00		
465	9.034e+00		
466	6.030e+00		
467	1.781e+00		
468	2.000e-01		
469	1.986e-01		
470	4.829e+00		
471	2.316e-02		
472	8.893e-01		

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores ≥ 0 do problema Bandm1

Coordenada y	Valor y (Dual)
5	9.121e-02
9	1.673e+00
14	4.646e-02
73	4.071e-01
77	2.465e+00
78	2.162e-02
79	2.420e-01
80	2.196e-01
82	2.692e-01
86	3.735e-01
102	8.000e-02
110	1.247e-01
117	1.632e+00
133	1.729e-01
147	1.269e+00
151	1.793e+00
159	3.011e-01
172	1.693e-01
173	4.829e-01
180	5.535e-02
200	1.000e-01
228	1.000e+00
230	1.326e-02
251	1.000e+00
256	1.000e-01
264	2.910e-03
270	2.324e-03
272	1.418e-02
273	1.257e-02
274	2.823e-06
280	2.668e-02
285	1.020e-02
287	2.285e-03
290	5.127e-03